

Newton-Vertex Obstructions for Polynomial-Graph Descents of the Gallagher Weighted Lift

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Abstract

We study polynomial-plane restrictions of the Gallagher weighted lift $F = (A, B, C) : \mathbb{A}^3 \rightarrow \mathbb{A}^3$. Let $\mathcal{U}(A, B, C)$ be its exact second-subduction coordinate. We prove that, on every polynomial graph $z = g(x, y)$,

$$\text{Jac}_{x,y}(\mathcal{U} + R, Q) \notin \mathbb{C}^\times$$

for every target polynomial $R(A, B, C)$ of degree at most two and every nonconstant, not necessarily homogeneous, $Q(A, B, C)$ of degree at most eight. More generally the conclusion holds in arbitrary target degree provided no two support exponents of Q differ by a nonzero multiple of $(5, -6, 4)$. This lattice direction is sharp: its primitive positive and negative parts give the first possible collision, A^5C^4 and B^6 , in degrees nine and six.

The proof uses the globally earliest Newton vertex, rather than an isolated homogeneous face. At a regular source boundary an exact seed gap protects either the leading Wronskian or its first nonmonomial deviation. At a pole boundary, cancellation at $u = 1$ is incompatible with the leading coefficient at $u = \infty$. The finite seed-support assertions are checked by exact rational arithmetic in the accompanying code.

This result excludes a broad descent architecture for a three-dimensional Keller counterexample. It neither proves nor disproves the two-dimensional Jacobian conjecture.

1 The weighted lift and the main theorem

The plane Jacobian conjecture and its standard reduction theory are surveyed in [2, 3]. The construction studied here is instead a descent problem arising from Gallagher’s three-dimensional weighted-lift family [1]; none of the arguments below changes the scope of the classical reductions. A targeted search conducted at the date of this draft found no statement matching Theorem 1.1; this is a cautious literature report, not a claim of priority.

Put

$$r(w) = w(w+1)(w-3)(w-4)(w-5)(w+4), \quad H(w) = -\frac{r(w)}{92},$$

and define

$$p(w) = H'(w) - H'(0), \quad q(w) = wH'(w) - H(w).$$

Their leading coefficients are

$$p_5 = -\frac{3}{46}, \quad q_6 = -\frac{5}{92}.$$

In source coordinates put

$$u = 1 + xy, \quad \gamma = 1 - \frac{57}{34}xy + x^2z, \quad w = u\gamma.$$

The weighted-lift coordinates have the exact presentations

$$A = \frac{q(w) + w\gamma}{x^2\gamma^2}, \quad B = \frac{p(w) + \gamma}{x\gamma}, \quad C = x\gamma. \quad (1)$$

The engineered divisibilities make the three expressions in (1) polynomials in x, y, z . Direct differentiation gives

$$\det \frac{\partial(A, B, C)}{\partial(x, y, z)} = 1.$$

This identity is checked from the explicit seed by `verify_weighted_lift_degree_six_graph_projection_no_go.py` and is also part of Gallagher's source construction [1].

We record the target polynomial used as the first coordinate. Let

$$\begin{aligned} t_1 &= -\frac{8505}{18948593792}, & t_2 &= -\frac{79515}{151588750336}, \\ t_3 &= \frac{1587485}{303177500672}, & t_4 &= -\frac{1079635}{303177500672}, \end{aligned}$$

and set

$$\begin{aligned} T &= p_5^6 A^5 C^4 - q_6^5 B^6 + t_1 A^4 B C^3 + t_2 A^3 B^2 C^2 \\ &\quad + t_3 A^2 B^3 C + t_4 A B^4. \end{aligned} \quad (2)$$

Write

$$\lambda_0 = \frac{258473511}{31221670147323559936}$$

and

$$\begin{aligned} s_0 &= -\frac{10204102899}{11489574614215070056448}, \\ s_1 &= \frac{5559334317}{249773361178588479488}, \\ s_2 &= -\frac{1028649031155}{11489574614215070056448}, \\ s_3 &= \frac{16337519127}{359049206694220939264}. \end{aligned}$$

The exact second subduction is

$$\mathcal{U} = p_5^5 C T - \lambda_0 B^5 + s_0 A^4 C^3 + s_1 A^3 B C^2 + s_2 A^2 B^2 C + s_3 A B^3. \quad (3)$$

Adding a constant to Q does not change its Jacobian with any first coordinate. We therefore replace Q by $Q - Q(0)$ throughout. For a polynomial

$$Q - Q(0) = \sum_{\substack{a,b,c \geq 0 \\ a+b+c > 0}} \lambda_{abc} A^a B^b C^c,$$

its support is the set of exponent triples with $\lambda_{abc} \neq 0$. Call the support *cuspid-free* if no two of its elements differ by a nonzero multiple of $(5, -6, 4)$.

Theorem 1.1 (Nonhomogeneous Newton-vertex obstruction). *Let $g \in \mathbb{C}[x, y]$, and restrict (1) to the graph $z = g(x, y)$. If $Q \in \mathbb{C}[A, B, C] \setminus \mathbb{C}$ has cuspid-free support and $\deg R \leq 2$, then*

$$\text{Jac}_{x,y}(\mathcal{U}(A, B, C) + R(A, B, C), Q(A, B, C)) \notin \mathbb{C}^\times. \quad (4)$$

Corollary 1.2 (Automatic cusp-free range through degree eight). *The conclusion of Theorem 1.1 holds for every nonconstant Q of total target degree at most eight. No homogeneity assumption is imposed.*

Corollary 1.3 (Homogeneous targets). *The conclusion of Theorem 1.1 holds for every nonzero homogeneous target $Q_n(A, B, C)$, in every degree n , and every $\deg R \leq 2$.*

2 Exact boundary templates

After restricting to $z = g(x, y)$, substitute

$$y = \frac{u-1}{x}.$$

Then

$$\gamma = 1 - \frac{57}{34}(u-1) + x^2 g\left(x, \frac{u-1}{x}\right) = \sum_{j=\rho}^M x^j f_j(u), \quad f_\rho \neq 0. \quad (5)$$

Necessarily $\rho \leq 0$. If there is no negative power, the coefficient of x^0 takes the value 1 at $u = 1$.

Let $\mathcal{N}(w, \gamma)$ be the numerator obtained from (3) after using (1) and clearing $x^5 \gamma^5$. Define

$$\begin{aligned} A(u, \gamma) &= \frac{q(u\gamma) + u\gamma^2}{\gamma^2}, \\ B(u, \gamma) &= \frac{p(u\gamma) + \gamma}{\gamma}, \\ H(u, \gamma) &= \frac{\mathcal{N}(u\gamma, \gamma)}{\gamma^5}. \end{aligned} \quad (6)$$

Thus

$$A = x^{-2}A, \quad B = x^{-1}B, \quad C = x\gamma, \quad \mathcal{U} = x^{-5}H. \quad (7)$$

Lemma 2.1 (Exact seed support). *The templates in (6) are polynomials and have unique top γ -degree terms*

$$\begin{aligned} A &= q_6 u^6 \gamma^4 + \text{terms of lower } \gamma\text{-degree}, \\ B &= p_5 u^5 \gamma^4 + \text{terms of lower } \gamma\text{-degree}, \\ H &= \theta u^{20} \gamma^{17} + \text{terms of lower } \gamma\text{-degree}, \end{aligned} \quad (8)$$

where

$$\theta = -\frac{36905625}{6077984970919772059860992} \neq 0.$$

If $\rho = 0$, put $f = f_0$ and $L = \deg f$. Then $L \geq 1$, and every non-top seed term in (8), after $\gamma = f$, loses at least $L + 1$ in u -degree. If $\rho < 0$, every non-top seed term has strictly later x -order.

Proof. The assertions follow by exact expansion of the degree-six seed and the two target subductions (2)–(3). For A, B , every non-top term loses at least one u -power and one γ -power. For H , the only additional edge possibility loses two γ -powers and no u -power. The regular loss is therefore at least $\min\{L + 1, 2L\} = L + 1$. When $\rho < 0$, losing a γ -power strictly raises the x -order.

The accompanying verifier expands all 77 nonzero terms of $\mathcal{N}$ over $\mathbb{Q}$, checks the divisibilities and support inequalities, and checks the displayed value of θ . No floating-point calculation is used. $\square$

We repeatedly use the elementary identity

$$\text{Jac}_{x,y}(x^\alpha H(u), x^\beta V(u)) = x^{\alpha+\beta}(\alpha H V' - \beta H' V), \quad (9)$$

which follows from $u = 1 + xy$.

3 Regular boundaries

Assume $\rho = 0$. Then

$$f(1) = 1, \quad f'(1) = -\frac{57}{34}, \quad L = \deg f \geq 1. \quad (10)$$

For an exponent triple $m = (a, b, c)$ define

$$\beta = -2a - b + c, \quad h = 6a + 5b, \quad q = 4a + 4b + c. \quad (11)$$

Notice that $q = h + \beta$. The top boundary term of $A^a B^b C^c$ is

$$q_6^a p_5^b x^\beta u^h f^q. \quad (12)$$

Let β_0 be the minimum β in $\text{Supp } Q$, and let h_0 be the maximum h among terms with $\beta = \beta_0$. The exponent triples at this vertex all have the same $q = h_0 + \beta_0$. Their aggregate coefficient is

$$\Lambda_0 = \sum_{\substack{\lambda_{abc} \neq 0 \\ \beta = \beta_0, h = h_0}} \lambda_{abc} q_6^a p_5^b. \quad (13)$$

Proposition 3.1 (Regular vertex). *If $\Lambda_0 \neq 0$, then $\text{Jac}(\mathcal{U}, Q) \notin \mathbb{C}^\times$.*

Proof. On a fixed β -face the u -degree of (12) is

$$h + qL = h + (h + \beta)L. \quad (14)$$

Decreasing h by $r \geq 1$ therefore costs exactly $r(L + 1)$. Lemma 2.1 gives the same protected gap for non-top seed terms. Consequently (13) supplies a unique nonzero aggregate at the globally earliest Jacobian order x^{β_0-5} .

For a vertex exponent triple, the coefficient Wronskian is

$$\mathcal{W} = -5HV' - \beta_0 H'V. \quad (15)$$

Its leading coefficient is a nonzero seed-and-target scalar times

$$B_* L + C_*, \quad (16)$$

where

$$B_* = -5q - 17\beta_0 = 14a - 3b - 22c, \quad C_* = -5h - 20\beta_0 = 10a - 5b - 20c. \quad (17)$$

If (16) is nonzero, then $\mathcal{W} \neq 0$.

Suppose (16) vanishes. The identity

$$17C_* - 15B_* = -10q \quad (18)$$

implies $B_* \neq 0$. The jets (10) rule out $f = du^L$, so

$$f = du^L + eu^{L-\delta} + \dots, \quad de \neq 0, \quad 1 \leq \delta \leq L. \quad (19)$$

After the leading cancellation, the first surviving coefficient in (15) is a nonzero multiple of

$$-B_*\delta e. \quad (20)$$

It occurs at degree drop at most L , before any other target vertex or non-top seed term can enter.

If $\beta_0 - 5 \neq 0$, the nonzero earliest x -term already contradicts a constant Jacobian. If $\beta_0 - 5 = 0$, the surviving coefficient still has positive u -degree. Before cancellation its degree is

$$20 + 17L + h_0 + qL - 1 > 0,$$

and (20) lowers it by only $\delta \leq L$. Thus it cannot be a nonzero constant. $\square$

4 Pole boundaries

Assume $\rho < 0$ and put $f = f_\rho$. A graph monomial $x^r y^s$ can contribute to f only when $r+2-s = \rho$; hence

$$f(u) = \sum_r c_r (u-1)^{r+2-\rho}.$$

Consequently

$$\nu := \text{ord}_{u=1} f \geq 2 - \rho, \quad L := \deg f \geq 2 - \rho, \quad L + \rho + 1 \geq 3. \quad (21)$$

The leading boundary terms are

$$\begin{aligned} \mathcal{U} &= x^\alpha \theta u^{20} f^{17} + \dots, & \alpha &= -5 + 17\rho, \\ A &= x^{-2+4\rho} q_6 u^6 f^4 + \dots, \\ B &= x^{-1+4\rho} p_5 u^5 f^4 + \dots, \\ C &= x^{1+\rho} f + \dots. \end{aligned} \quad (22)$$

For a target exponent triple retain q from (11) and put

$$E = \beta + \rho q. \quad (23)$$

Let E_0 be the minimum E in the support. At the globally earliest face, group the top target terms by q :

$$\Lambda_q = \sum_{\substack{\lambda_{abc} \neq 0 \\ E=E_0, q(a,b,c)=q}} \lambda_{abc} q_6^a p_5^b. \quad (24)$$

Let $q_{\min}$ and $q_{\max}$ be the smallest and largest indices with $\Lambda_q \neq 0$.

Proposition 4.1 (Polar endpoints). *If some $\Lambda_q \neq 0$, then $\text{Jac}(\mathcal{U}, Q) \notin \mathbb{C}^\times$.*

Proof. At fixed E_0 , the q -group is a nonzero multiple of

$$u^s f^q, \quad s = (1 + \rho)q - E_0. \quad (25)$$

At $u = 1$, the $q_{\min}$ -group uniquely has least vanishing order. Its f' -coefficient in the Wronskian is

$$\alpha q_{\min} - 17E_0. \quad (26)$$

If this is nonzero, the resulting term has order

$$(17 + q_{\min})\nu - 1 > 0 \quad (27)$$

at $u = 1$, and every larger q has strictly greater order.

Suppose instead that (26) vanishes, so

$$E_0 = \frac{\alpha q_{\min}}{17}. \quad (28)$$

At infinity the $q_{\max}$ -group uniquely dominates because $L + \rho + 1 > 0$. Its leading coefficient is

$$\alpha \left((L + \rho + 1)(q_{\max} - q_{\min}) + \frac{2q_{\min}}{17} \right), \quad (29)$$

which is nonzero. Its degree is

$$20 + 17L + (L + \rho + 1)q_{\max} - E_0 - 1 > 0. \quad (30)$$

Here $E_0 \leq 0$: for $\rho \leq -1$, the coefficients of a, b, c in E are respectively $-2 + 4\rho, -1 + 4\rho, 1 + \rho$, all nonpositive. Every non-top seed term has later x -order by Lemma 2.1. Hence the earliest coefficient is nonzero and, if its x -order is zero, is still nonconstant by (27) or (30). $\square$

5 The cusp kernel and the degree-eight cutoff

Lemma 5.1 (Primitive collision direction). *Two exponent triples have the same pair (β, h) if and only if their difference is*

$$k(5, -6, 4), \quad k \in \mathbb{Z}. \quad (31)$$

For every fixed $\rho < 0$, equality of (E, q) has the same kernel.

Proof. The integer kernel of

$$\begin{pmatrix} -2 & -1 & 1 \\ 6 & 5 & 0 \end{pmatrix}$$

is generated primitively by $(5, -6, 4)$. Replacing the first row by $E = \beta + \rho q$ and the second by $q = (4, 4, 1)$ leaves the same kernel. $\square$

Proof of Theorem 1.1. At a regular boundary, cusp-freeness makes the vertex in (13) a single nonzero target coefficient. Proposition 3.1 applies. At a pole boundary, Lemma 5.1 makes every nonempty aggregate in (24) nonzero, so Proposition 4.1 applies.

It remains to include R . At a regular boundary, every nonconstant monomial of target degree at most two has x -order at least $-4 > -5 = \text{ord}_x \mathcal{U}$. At a pole boundary such a monomial has $\beta \geq -4$ and $q \leq 8$, whence

$$E_R - \alpha = (\beta + 5) + \rho(q - 17) > 0. \quad (32)$$

Thus $\text{Jac}(R, Q)$ occurs strictly after the protected earliest coefficient of $\text{Jac}(\mathcal{U}, Q)$. $\square$

Proof of Corollary 1.2. Suppose two distinct nonnegative exponent triples differ by $k(5, -6, 4)$, with $k > 0$ after exchanging them. The triple on the negative- B side has $b \geq 6k$, so its total target degree is at least $6k$. The other triple has total degree larger by $3k$, hence at least $9k$. Such a pair cannot occur when all monomials have degree at most eight. $\square$

Proof of Corollary 1.3. The total degree changes by $5 - 6 + 4 = 3$ along the primitive kernel. Distinct monomials in one homogeneous target therefore cannot differ by a nonzero multiple of (31). $\square$

6 Why degree nine is the first unresolved block

The primitive positive and negative parts of (31) give

$$A^5C^4 \quad \text{and} \quad B^6. \quad (33)$$

Their boundary top forms are proportional:

$$p_5^6 A_{\text{in}}^5 C_{\text{in}}^4 = q_6^5 B_{\text{in}}^6. \quad (34)$$

This is the first point at which the aggregate coefficient in the regular or polar proof can cancel.

The exact polynomial T in (2) subducts this relation, and (3) subducts the next relation $TC \sim B^5$. It is tempting to infer that $A, B, C, T, \mathcal{U}$ form a complete SAGBI basis, but this is false. The genuine next saturation

$$S_0 = q_6^5 BT - p_5 \lambda_0 A^5 C^3 \quad (35)$$

cancels its common initial monomial. Define

$$\begin{aligned} W = S_0 - & \frac{1289186932405}{367666387654882241806336} A^4 BC^2 \\ & + \frac{967724214885}{183833193827441120903168} A^3 B^2 C \\ & + \frac{280059055635}{367666387654882241806336} A^2 B^3. \end{aligned} \quad (36)$$

These are the three successive available monomial reductions on the same boundary face. Exact expansion shows that the regular-boundary Pareto frontier of W is

$$(-7, 26, 19), \quad (-7, 25, 20) \quad (37)$$

in (x, u, f) -exponents. Neither vector belongs to the semigroup generated by the five known initial vectors. A new generator is therefore required. This calculation prevents an unsupported finite-normal-form claim; it does not prove that the full SAGBI completion is infinite.

7 Computational reproducibility and scope

All exact computations use rational arithmetic. From the repository root, the principal checks are

```
.venv/bin/python \
  current_context/verify_weighted_lift_degree_six_graph_projection_no_go.py
.venv/bin/python \
  current_context/verify_weighted_lift_global_minimal_boundary_face_closure.py
.venv/bin/python \
  current_context/verify_weighted_lift_nonhomogeneous_newton_vertex_closure.py
.venv/bin/python \
  current_context/verify_weighted_lift_sagbi_saturation_next_generator_audit.py
```

The first script checks the three-dimensional Keller identity. The second reconstructs the seed from $r(w)$, builds (2)–(3), checks the exact numerator support and seed gaps, and verifies the boundary formulas. The third independently checks all derived Wronskian, endpoint, lattice, cutoff, and perturbation identities. The fourth checks the cautionary next saturation (35)–(37).

The archived code version for this manuscript is

[https://github.com/AtharvaVaidya/mathematics-research/tree/
weighted-lift-newton-obstructions-v1](https://github.com/AtharvaVaidya/mathematics-research/tree/weighted-lift-newton-obstructions-v1).

The theorem concerns polynomial graph embeddings into one specific three-dimensional Keller family and a specified first target coordinate. It does not cover arbitrary nonlinear first-coordinate perturbations, non-graph affine-plane embeddings, or arbitrary plane Keller maps. In particular it is not a proof of the plane Jacobian conjecture.

Disclosure

OpenAI Codex assisted with symbolic computation, proof search, adversarial review, and drafting. The named author is responsible for the claims. This manuscript should receive independent human line-by-line verification and a broader expert novelty review before submission.

References

- [1] A. Gallagher, *The Jacobian counterexample, explained*, <https://jacobianfun.org/jacobian-explained>, accessed 25 July 2026; archived construction, <https://doi.org/10.5281/zenodo.21479195>.
- [2] H. Bass, E. Connell, and D. Wright, The Jacobian conjecture: reduction of degree and formal expansion of the inverse, *Bull. Amer. Math. Soc.* **7** (1982), 287–330.
- [3] A. van den Essen, *Polynomial Automorphisms and the Jacobian Conjecture*, Progress in Mathematics 190, Birkhäuser, 2000.